

Database Systems

LAB#09



23K-2001

BCS-5J

Q1:

```
CREATE TABLE emp_audit (  
    audit_id NUMBER GENERATED ALWAYS AS IDENTITY,  
    emp_id NUMBER,  
    action_type VARCHAR2(20),  
    action_date DATE,  
    old_salary NUMBER,  
    new_salary NUMBER  
);  
/
```

```
CREATE OR REPLACE TRIGGER trg_emp_dml_log  
AFTER INSERT OR UPDATE OR DELETE ON employees  
FOR EACH ROW  
BEGIN  
    IF INSERTING THEN  
        INSERT INTO emp_audit(emp_id, action_type, action_date, new_salary)  
        VALUES(:NEW.employee_id, 'INSERT', SYSDATE, :NEW.salary);  
  
    ELSIF UPDATING THEN  
        INSERT INTO emp_audit(emp_id, action_type, action_date, old_salary, new_salary)  
        VALUES(:NEW.employee_id, 'UPDATE', SYSDATE, :OLD.salary, :NEW.salary);  
  
    ELSIF DELETING THEN  
        INSERT INTO emp_audit(emp_id, action_type, action_date, old_salary)  
        VALUES(:OLD.employee_id, 'DELETE', SYSDATE, :OLD.salary);  
    END IF;  
END;  
/
```

Q2:

```
CREATE OR REPLACE TRIGGER trg_orders_fk_check
BEFORE INSERT OR UPDATE ON orders
FOR EACH ROW
DECLARE
    v_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_count
    FROM customers
    WHERE customer_id = :NEW.customer_id;

    IF v_count = 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Customer does not exist.');
```

END IF;

```
END;
/
```

Q3:

```
CREATE OR REPLACE TRIGGER trg_update_timestamp
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    :NEW.last_modified := SYSDATE;
END;
/
```

Q4:

```
CREATE TABLE ddl_audit (  
  id NUMBER GENERATED ALWAYS AS IDENTITY,  
  username VARCHAR2(50),  
  operation VARCHAR2(50),  
  object_type VARCHAR2(30),  
  object_name VARCHAR2(100),  
  change_date DATE  
);  
/
```

```
CREATE OR REPLACE TRIGGER trg_ddl_log  
AFTER DDL ON SCHEMA  
BEGIN  
  INSERT INTO ddl_audit(username, operation, object_type, object_name, change_date)  
  VALUES(USER, ORA_SYSEVENT, ORA_DICT_OBJ_TYPE, ORA_DICT_OBJ_NAME,  
SYSDATE);  
END;  
/
```

Q5:

```
CREATE OR REPLACE TRIGGER trg_protect_employees  
BEFORE DROP OR ALTER ON SCHEMA  
BEGIN  
  IF ORA_DICT_OBJ_NAME = 'EMPLOYEES' THEN  
    RAISE_APPLICATION_ERROR(-20002, 'Modification of EMPLOYEES table is not  
allowed.');
```

```
  END IF;  
END;  
/
```

Q6:

```
CREATE OR REPLACE TRIGGER trg_block_tmp_tables
BEFORE CREATE ON SCHEMA
BEGIN
    IF ORA_DICT_OBJ_NAME LIKE 'TMP_%' THEN
        RAISE_APPLICATION_ERROR(-20003, 'Tables starting with TMP_ are not allowed.');
```

END IF;

```
END;
/
```

Q7:

```
CREATE TABLE login_audit (
    username VARCHAR2(40),
    login_time DATE,
    machine VARCHAR2(100)
);
/

CREATE OR REPLACE TRIGGER trg_log_login
AFTER LOGON ON DATABASE
BEGIN
    INSERT INTO login_audit(username, login_time, machine)
    VALUES(USER, SYSDATE, SYS_CONTEXT('USERENV','HOST'));
END;
/
```

Q8:

```
CREATE OR REPLACE TRIGGER trg_notify_dba
AFTER LOGON ON DATABASE
BEGIN
    IF USER IN ('ADMIN_USER') THEN
        UTL_MAIL.send(
            sender    => 'system@db.com',
            recipients => 'dba@db.com',
            subject   => 'Privileged Login',
            message   => USER || ' logged in at ' || SYSDATE
        );
    END IF;
END;
/
```

Q9:

```
CREATE OR REPLACE TRIGGER trg_set_timezone
AFTER LOGON ON DATABASE
BEGIN
    EXECUTE IMMEDIATE 'ALTER SESSION SET TIME_ZONE = "+05:00" ';
END;
/
```

Q10:

```
CREATE OR REPLACE VIEW v_emp_dept AS
SELECT e.emp_id, e.emp_name, d.dept_id, d.dept_name
FROM employees e
JOIN departments d USING (dept_id);
/
```

```
CREATE OR REPLACE TRIGGER trg_v_emp_dept_insert
INSTEAD OF INSERT ON v_emp_dept
BEGIN
    INSERT INTO departments(dept_id, dept_name)
    VALUES(:NEW.dept_id, :NEW.dept_name);

    INSERT INTO employees(emp_id, emp_name, dept_id)
    VALUES(:NEW.emp_id, :NEW.emp_name, :NEW.dept_id);
END;
/
```

Q11:

```
CREATE OR REPLACE VIEW v_emp_salary AS
SELECT emp_id, salary, salary * 12 AS annual_salary
FROM employees;
/
```

```
CREATE OR REPLACE TRIGGER trg_v_emp_salary_update
INSTEAD OF UPDATE ON v_emp_salary
BEGIN
    UPDATE employees
    SET salary = :NEW.annual_salary / 12
    WHERE emp_id = :OLD.emp_id;
END;
/
```

Q12:

```
CREATE OR REPLACE VIEW v_cust_order AS
SELECT c.customer_id, c.customer_name, o.order_id, o.order_amount
FROM customers c
LEFT JOIN orders o USING(customer_id);
/
```

```
CREATE OR REPLACE TRIGGER trg_v_cust_order_insert
INSTEAD OF INSERT ON v_cust_order
BEGIN
    INSERT INTO customers(customer_id, customer_name)
    VALUES(:NEW.customer_id, :NEW.customer_name);

    INSERT INTO orders(order_id, order_amount, customer_id)
    VALUES(:NEW.order_id, :NEW.order_amount, :NEW.customer_id);
END;
/
```